

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	SWTID-2026-4683
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:
This document evaluates the quality of the EduGenie codebase in terms of its structure, readability, and reusability. The assessment follows the checklist and table organization shown in the provided template.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the Python, HTML and CSS code.	Yes	Use consistent Python indentation and consistent HTML/CSS formatting.
2	Proper File Structure	Files and folders are logically organized by application responsibility.	Yes	Separate frontend templates, static assets, backend modules and configuration where applicable.
3	Meaningful Variable Names	Variables reflect their purpose clearly.	Yes	Use descriptive names such as question, topic, summary, quiz and learning_path.
4	Function / Method Names	Functions are descriptively named.	Yes	Use names that indicate the feature or operation being performed.
5	Code Comments	Inline and block comments explain important logic.	Partial	Comments should focus on non-obvious AI integration, model loading and API behavior.
6	Modular Design	Code is split into reusable functions/modules.	Yes	Learning capabilities such as Q&A;, quiz, summarization and recommendations can be kept modular.
7	No Redundant Code	Duplicate or unused code is removed.	Yes	Shared request/response logic should be reused rather than duplicated across endpoints.
8	Error Handling	Exceptions and errors are handled gracefully.	Yes	API failures, invalid input and model/service errors should return clear user-facing messages.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Q&A; / Prompt Processing Module	Python, FastAPI, Gemini API	Question answering and shared learner-query processing	High
2	Concept Explanation Module	Python, LaMini-Flan-T5-783M	Concept explanation requests and educational responses	High
3	Quiz Generation Module	Python, FastAPI, Gemini API	Quiz creation for different topics and learning content	High
4	Summarization Module	Python, FastAPI, Gemini API	Summarizing educational text supplied by learners	High
5	Learning Recommendation Module	Python, FastAPI, Gemini API	Generating learning paths and topic progression	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Modular FastAPI structure with separate learning capabilities and a clear frontend/backend separation.
Readability	4	Descriptive feature names and straightforward API flow support readability.
Reusability	4	Feature modules and shared API/model integration logic can be reused across learning tasks.
Documentation / Comments	3	Basic documentation is appropriate; additional comments can be added around non-obvious model and API integration.

Overall Score	4 / 5	Good overall structure for a modular AI learning assistant, with documentation and consistency as areas for continued refinement.
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